

Curcumin sensitizes tumor necrosis factor-related apoptosis-inducing ligand (TRAIL)-induced apoptosis through reactive oxygen species-mediated upre...

Curcumin exhibits anti-inflammatory and antitumor activities. Although its functional mechanism has not been elucidated so far, numerous studies have shown that curcumin induces apoptosis in cancer cells. In the present study, we show that subtoxic concentrations of curcumin sensitize human renal cancer cells to the tumor necrosis factor-related apoptosis inducing ligand (TRAIL)-mediated apoptosis. This apoptosis induced by the combination of curcumin and TRAIL is not interrupted by Bcl-2 overexpression. We found that treatment with curcumin significantly induces death receptor 5 (DR5) expression both at its mRNA and protein levels, accompanying the generation of the reactive oxygen species (ROS). Not only the pretreatment with N-acetylcystine but also the ectopic expression of peroxiredoxin II, an antioxidative protein, dramatically inhibited the apoptosis induced by curcumin and TRAIL in combination, blocking the curcumin-mediated DR5 upregulation. Taken together, the present study demonstrates that curcumin enhances TRAIL-induced apoptosis by ROS-mediated DR5 upregulation.